

A Deep Learning–Based Approach for Empirical Modeling of Single-Point Wave Spectra in Open Oceans

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ABSTRACT: Directional wave spectra are of importance for numerous practical applications such as seafaring and ocean engineering. The wave spectral densities at a certain point in the open ocean are significantly correlated to the local wind field and historical remote wind field. This feature can be used to predict the wave spectrum at that point using the wind field. In this study, a convolutional neural network (CNN) model was established to estimate wave spectra at a target point using the wind field from the ERA5 dataset. A geospatial range where the wind could impact the target point was selected, and then the historical wind field data within the range were analyzed to extract the nonlinear quantitative relationships between wind fields and wave spectra. For the spectral densities at a given direction, the wind data along the direction where waves come from were used as the input of the CNN. The model was trained to minimize the mean square error between the CNN-predicted and ERA5 reanalysis spectral density. The data structure of the wind input is reorganized into a polar grid centered on the target point to make the model applicable to different open-ocean locations worldwide. The results show that the model can predict well the wave spectrum shapes and integral wave parameters. The model allows for the prediction of single-point wave spectra in the open ocean with low computational cost and can be helpful for the study of spectral wave climate.

SIGNIFICANCE STATEMENT: The directional wave spectra (DWS) describe the distribution of wave energy among different frequencies and directions. They are useful for many marine practical applications. Usually, DWS are modeled using numerical wave models (NWMs) based on wave action balance differential equations. Although contemporary NWMs perform well after years of development, their computational costs are relatively high. The fast-developed artificial intelligence (AI) might provide an alternative solution to this task. In this study, convolutional neural networks are used to model the DWS at some selected points in the open ocean. By “learning” from NWM data, AI can effectively simulate single-point DWS in open oceans with low computational cost, which can serve as a faster data-driven surrogate model in related applications.

KEYWORDS: Wind waves; Oceanic waves; Neural networks; Deep learning; Regression

1. Introduction

Wind-generated surface gravity waves are a very common phenomenon on the sea surface and play an important role in many human activities such as maritime transport, fisheries, and ocean engineering. The study of wave conditions is therefore of great social and scientific importance. Traditionally, integral ocean wave parameters, such as significant wave height (SWH), mean wave direction (MWD), and mean wave period (MWP), are used to describe wave states. However, because the wave at most locations in the ocean is the superimposition of wind-seas generated locally and swells generated from distant regions, these integral wave parameters provide only a limited description of the wave field and can be misleading in

sea states with multiple wave components. A more comprehensive way of describing the wave field is the wave spectrum, which can characterize information about waves from different origins (e.g., Holthuijsen 2007).

Numerical wave models (NWMs), such as “WAVEWATCH III” (Tolman and WAVEWATCH III Development Group 2014) and Simulating Waves Nearshore (SWAN; Tolman 1991; Booij et al. 1997), are the most widely used tools to simulate the wave spectrum. After years of development and iteration, NWMs can obtain good results of wave parameters with high-quality forcing (e.g., The WISE Group 2007; Alday et al. 2021; Liu et al. 2021). Meanwhile, the computational costs of NWMs are still relatively high, especially for the setting with high spatial, temporal, and spectral resolutions. Some numerical effects, such as the discretization of frequencies and directions during wave propagation [so-called garden sprinkler effect (GSE)] (Tolman 2002) and the numerical approximation of the solution to the quadruplet wave–wave interactions (Hasselmann et al. 1985), also affect the performance of the model output. Besides, even if one only wants the wave state at a selected location, it is still necessary to model the wave of the whole basin when an NWM is used.

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Waves are generated by either local wind or remote wind, thus, the wave spectral densities at a certain point in the ocean are significantly correlated to local wind field and historical remote wind field. This has been used in swell prediction before the 1950s (e.g., Sverdrup and Munk 1946; Barber and Ursell 1948). Using the assumption that deep water waves travel along great circle paths, Pérez et al. (2014) proposed a method to conveniently determine the regions with potential impacts on the wave state at a given location in the ocean by shorting great circles at different directions. The wave energy flux and travel time from different regions to the selected location can also be estimated using their method. Jiang and Mu (2019) proposed that each frequency and direction of the wave spectrum at a given location can be approximately correlated with the wind field in a geographical area, which is physically due to the fact that the local waves are made up of waves generated by wind from this geographical area. This implies that it is feasible to reconstruct the wave spectrum using wind field data. This statistical correlation between the wave spectrum and the wind field offers the possibility of establishing a purely empirical model to estimate the wave spectrum at a given location using local and remote wind fields.

Although the wind is the main reason for the generation of waves, the relation between local wave spectra and the global wind is very complex because this relation involves many nonlinear physical processes such as the generation, propagation, dissipation, and interactions of wave energy. Some previous studies used dimensionality reduction methods, such as principal component analysis and clustering algorithm, to identify the complex relations between the pressure synoptic pattern and single-point spectral energy distribution (e.g., Cagigal et al. 2021; Espejo et al. 2014; Hegermiller et al. 2017; Rueda et al. 2017), demonstrating the feasibility of such purely empirical method of wave spectrum modeling.

The rapid development of artificial intelligence (AI) technology might provide an alternative solution to this task. AI is able to “learn” the features and correlations between inputs and outputs from a large amount of data by supervised training. Therefore, they are suited for the problem that we know there are some causal relationships between inputs and outputs, but the accurate analytical physical model is too sophisticated to establish. AI has been recently widely used in many aspects of oceanography (e.g., Ham et al. 2019; Jiang 2022; Zheng et al. 2020), including modeling and prediction of ocean waves (e.g., James et al. 2018; O'Donncha et al. 2018). These studies validate the feasibility of using neural networks for wave modeling, but their predicants are mainly integral wave parameters such as SWH, MWP, and MWD instead of directional wave spectra. Therefore, this study aims to establish an empirical wave model based on convolutional neural networks (CNNs) to estimate single-point directional wave spectra using wind field data. We hope such an empirical wave model can “re-learn” the pattern of wind-wave spectral growth (including nonlinear wave-wave interactions) with varying space-time winds and the propagation of swell spectra all at a once from the data. It is noted that the aim of such AI models is not to replace physics-based numerical wave spectral models, but to develop a faster data-driven surrogate model.

The structure of this paper is as follows. Section 2 presents the data of directional wave spectra and sea surface wind fields used in this study, as well as the preprocessing of the data. The structure and the training method of the CNN models are also introduced in section 2. The main results are presented in section 3 along with a brief discussion about this methodology. In this section, the applicability of this model in the open ocean worldwide is demonstrated, followed by the concluding remarks in section 4.

2. Data and method

a. Wind field and wave spectrum data

The data of 10-m wind fields and directional wave spectra from the ERA5 dataset were used in this study. ERA5 is the fifth major global reanalysis produced by the European Centre for Medium-Range Weather Forecasts, covering the period from 1950 to present and providing hourly data on many atmospheric and wave parameters with a resolution of 31 km (Hersbach et al. 2020). The dataset is available from the Climate Data Store with a preinterpolated spatial resolution of up to $0.25^\circ \times 0.25^\circ$ for wind fields and $0.5^\circ \times 0.5^\circ$ for directional wave spectra. The wave spectrum data of ERA5 consist of 24 directional bins with 15° spacing and 30 frequency bins increasing exponentially from 0.0345 to 0.5473 Hz.

This study used the directional wave spectra from an NWM output instead of those from buoy observations for two reasons:

- 1) The directional wave spectra from buoy measurements are estimated from the so-called first-five Fourier coefficients using reconstruction methods such as the maximum entropy method (Earle et al. 1999). Although these methods are widely applied in ocean engineering, they have problems such as reducing the directional spread and generating spurious peaks. Besides, the reconstructed directional wave spectra from buoy measurements are often noisy with respect to spectral densities at different frequency-direction bins (e.g., Jiang et al. 2022). Therefore, they are not suited to serve as the training target of the CNN models.
- 2) The training of a CNN model needs a large amount of wave spectrum data, and NWMs can theoretically generate as much data as needed to train the model. Also, contemporary NWMs can give an estimation of directional ocean wave spectra that are consistent with those measured by buoys, especially in the open ocean (e.g., Alday et al. 2021; Liu et al. 2021). The aim of this study is to develop a surrogate for the NWMs to save computational cost; thus, the spectral output of NWMs as the training target is an acceptable option.

b. Processing of wind input

A location at 5°N , 120°W (hereinafter point A) is selected here as an example to explain the procedure of data processing. This point is located at both the Pacific “swell pool” (Chen et al. 2002) and “crossing swell pool” (Jiang et al. 2017) where the westerlies-generated swell systems from both hemispheres and the trade wind-generated wave systems (can be

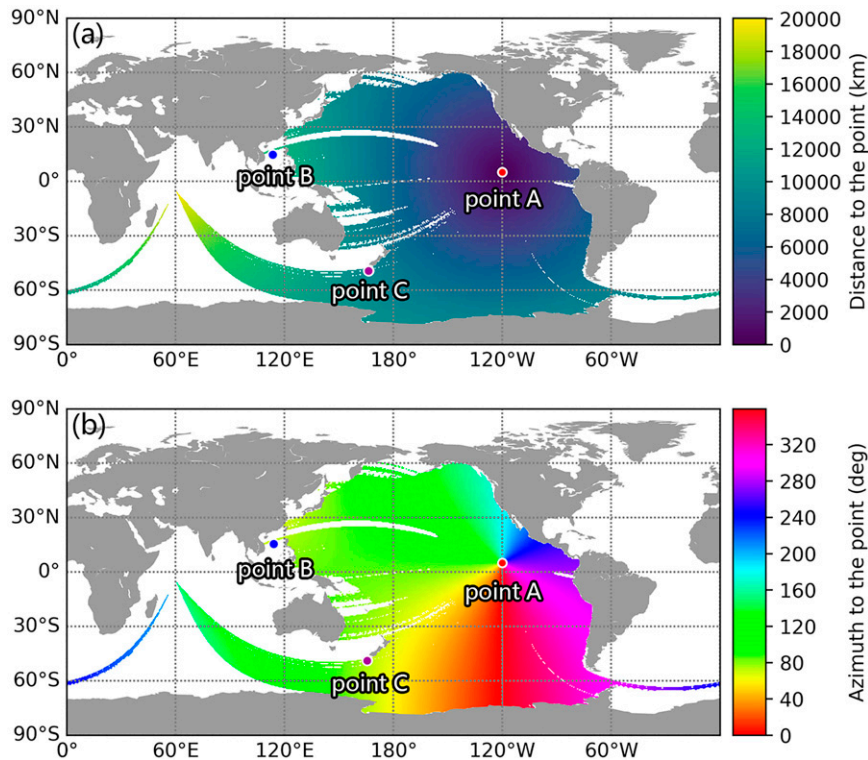


FIG. 1. Regions where generated waves can have potential impacts on waves at point A (the red point) considering only the great-circle propagation and land blocking of waves: the (a) propagating distances and (b) azimuths from different locations in the ocean to point A. For the convention of directions, 0° and 90° refer to northward and eastward propagation, respectively. point B and point C in this study are also marked in blue and purple, respectively.

either wind-seas or swells) often coexist at the same time (Jiang and Mu 2019). Therefore, the wind information from the remote westerlies is necessary for the prediction of wave spectra at this location. The regions where wind fields have potential impacts on the wave conditions at the selected location are determined using the spheric geometrical optical model (e.g., Snodgrass et al. 1966; Barber and Ursell 1948) ignoring the effects of refraction, reflection, and diffraction in the wave propagation, which has been used in Pérez et al. (2014) and Jiang and Mu (2019). Figures 1a and 1b show the propagating distances and azimuths from different locations in the ocean to point A, after considering the land blocking of waves. It is noted that the directions in this paper are in the “ocean convention,” meaning 0° represents northward and 90° represents eastward. The wind data outside these potentially impacting regions are not used in the empirical modeling of wave spectra at point A.

Waves are directly generated by local winds, so winds near the target point can have a large impact on the wave spectra, especially at the high-frequency part. The impact of remote winds on the target point is through the propagation of swells. High-frequency energy is dissipated more quickly than low-frequency energy during wave propagation, and low-frequency energy gradually dominates as the propagation distance increases. Besides, because of the frequency dispersion, angular spread, and energy dissipation,

the wave energy attenuates significantly before reaching the target point. Therefore, remote winds mainly impact the low-frequency wave spectra at the target point, and the impact is smaller than that of local winds.

Therefore, the detailed pattern of local wind (as well as the wind near the selected point) is very important for the modeling of wave spectra, but the wind information in the distant areas can be represented using a relatively coarse-resolution wind field. Taking into account the balance between accuracy and computational cost, the wind field data are reorganized into a polar grid centered on the target point.

An illustration of the polar grid used in this study is shown in Fig. 2. The grid is divided into 120 directions around point A with 3° spacing. Because the detailed pattern of the wind field is more important close to the target point, the “longitude line” of the grid is divided with an initial grid length of 50 km and an exponentially increasing factor of 1.05. When the length of the grid along longitude lines exceeds 400 km ($50 \text{ km} \times 1.05^{43} \approx 407.5 \text{ km}$), the subsequent grids stop growing and are all set to 400 km, which is to avoid the grid becoming too coarse in the far field. The most distant grid point is set to 10 000 km, because the energy of swells coming from farther distances will decay to a negligible level (e.g., Ardhuin et al. 2009) and have almost no impact on the target point. Using this method, we can obtain a grid with the grid points denser near the target point and

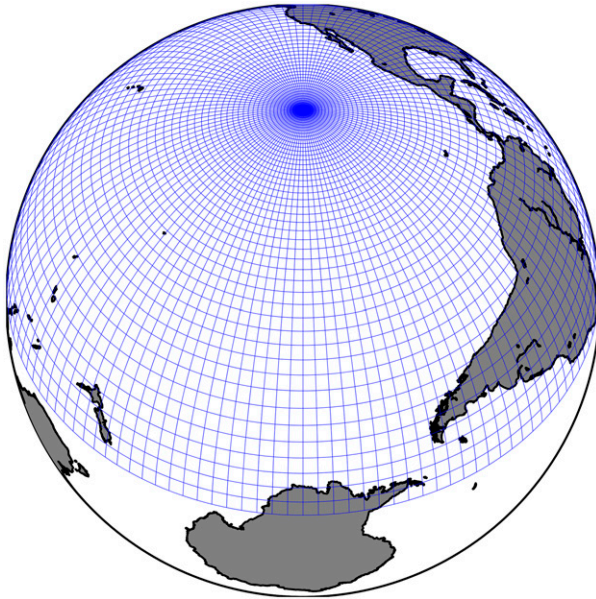


FIG. 2. The polar grid for wind field used in this study centered on point A.

becoming coarse (but not too coarse) with the increase of distances.

The original wind fields from ERA5 were interpolated into the new polar grid using the nearest-neighbor method. Because of the great circle propagation of waves, winds blowing in the opposite direction toward point A cannot produce energy that affects point A. Jiang and Mu (2019) showed that the local wave energy at a given direction in the target location is significantly positively correlated with the projection of wind vector toward the target location. Such a correlation can be the statistical basis for the prediction of spectral density. The definition of this wind projection is

$$U_{\text{proj}} = \max[0, U \cos(\theta_U - \theta_A)\delta], \quad (1)$$

where U is the 10-m wind speed, θ_U is the wind direction, θ_A is the azimuth from a point to the target point along the great circle, and δ is the land blocking factor that takes the value 0 or 1, which means the location is with/without land blocks toward the target point. By further doing a correlation analysis with the wave spectrum, a stronger correlation was found when replacing the U in Eq. (1) with U^2 so that U_{proj}^2 is also used as the input of the model:

$$U_{\text{proj}}^2 = \max[0, U^2 \cos(\theta_U - \theta_A)\delta]. \quad (2)$$

These two input terms are sufficient to fully describe the wind vector in a given grid; thus, no other input features were selected.

According to Jiang and Mu (2019), the wave spectral densities at a given direction at point A are mainly impacted by the wind field along the direction where the waves come from. For the prediction of the spectral densities at the given direction, only the wind information in the direction where the

waves come from is needed as the input of the empirical model. After using such a polar grid and wind field data reorganization, the empirical model for wave spectrum prediction can be established using the wind field from five direction sectors ($5 \times 3^\circ = 15^\circ$) as the input and the spectral density in the corresponding direction as the output. This can effectively reduce the complexity of the model as explained in the next section.

c. The CNN model for wave spectra

In this study, an empirical spectral wave model based on deep learning is constructed. The CNN is currently one of the dominant algorithms in the field of AI. It has been shown that CNNs are able to extract useful information from complex data more effectively and are now being used in many research areas. They can acquire different levels of representation through different layers, learning more abstract representations as the layers are added. In addition to this, it also needs to be backed up by a large amount of good-quality training data. In this way, CNNs can extract useful information from complex, high-dimensional data (Bengio et al. 2021; LeCun et al. 2015). Therefore, CNNs were also selected as the algorithm to establish the model in this study.

The architecture of the CNN-based empirical spectral wave model is shown in Fig. 3. The U_{proj} and U_{proj}^2 from the past 10 days in 15° sectors of the polar grid in Fig. 2 with a temporal resolution of 3 h are used as the input of the CNN model. The reason for using the wind information of the past 10 days is that it takes up to ~ 10 days for the swells to travel across the Pacific basin. Each of the input samples is a three-dimensional matrix with a shape of $10 \times 80 \times 50$ (2 parameters $\times$ 5 direction sectors, 10 days $\times$ 8 fields day $^{-1}$, and 50 polar grid points in each 3° sector). The target output of the CNN model is the spectral densities corresponding to the direction of the 15° sector at the target location (e.g., point A). By predicting the spectra at different directions separately using the corresponding sector in the polar grid, the size of the input vector was reduced to 1/24 while the number of samples was increased for 24 times, which largely reduced the complexity of model training. That is also one reason for selecting the polar grid. The separation of directions might lead to the missing of some wind field information and thus introduce some errors, but the model will become too complex to train if the information of all directions is used together as the model input. The polar grids with lower and higher resolutions were also tested. Halving the resolution will give a significantly worse result while doubling the resolution does not give a significantly better result.

A total of 84 000 samples (3500 spectra $\times$ 24 spectral directions) from data of 20 years (1994–2013) were used as the training set, and 21 000 samples (875 spectra $\times$ 24 spectral directions) from data of 5 years (2014–18) were used as the validation set. The samples were selected using an interval of 48 h to enhance the independence of each spectrum. Note that the samples are still hourly spectra instead of 48-h mean spectra, and the selection of 48 h is arbitrary, since this is simply to remove some probably redundant data in the training dataset to speed up the training process, because two spectra with short

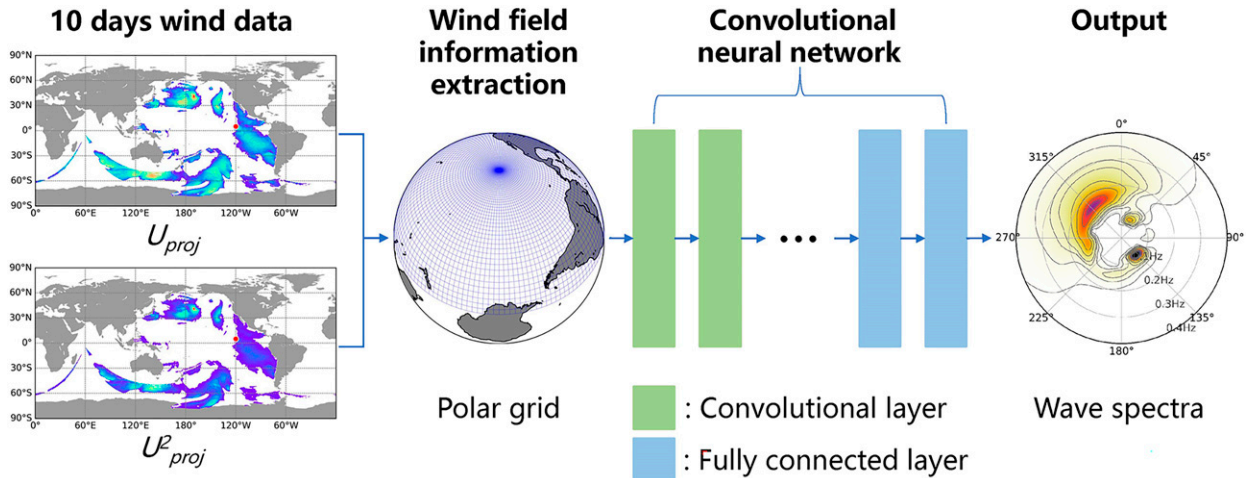


FIG. 3. The architecture of the CNN-based empirical spectral wave model.

time interval will be similar. After this procedure, only 175 spectra were obtained each year.

An independent model is trained for the spectral density of each frequency so that 30 parallel models were obtained to predict the spectral densities at a given direction. Because we want to make the model “purely data driven,” no restrictions were made on the shape of spectra. For each of the parallel models, eight convolutional layers and four fully connected layers are used. The size of convolutional kernels is 3×3 . The eight-convolution layer has 64, 128, 512, 256, 128, 64, 16, and 8 convolution kernels, respectively. After each convolutional layer, batch normalization was performed, followed by “max pooling” operation after rectified linear unit (ReLU) activation function processing. The fully connected layers consist of an input layer (with 120 neurons), two hidden layers (with 32 and 8 neurons, respectively), and an output layer (with 1 neuron). The ReLU activation function is also applied after each fully connected layer. The CNN was trained to minimize the MSE between the output and the target (predicted and referenced spectral density) using the Adam optimizer with a batch size of 256. The learning rate (initially set to 0.001) was decreased by 50% if the RMSE of the training set did not decrease for five epochs, and the training process stopped when the RMSE of the validation set did not decrease for 10 epochs. The 3×3 kernels, the max pooling, ReLU, and Adam optimizer are all widely used settings in CNN. Increasing the kernel size and changing these settings do not give a better result. The structure of this CNN (including the number of layers, number of kernels, and number of neurons) is determined based on experience and some trivial experiments. According to these experiments, increasing the numbers of layers, kernels, and neurons does not give better results on the testing set.

To optimize the training process, we adopted an approach of training from high to low frequencies. The model parameters should be similar for similar frequencies because the values of spectral densities are often close for two adjacent frequency bins. Therefore, the model for the highest frequency

was trained using an initial setting of random parameters while the model for the next frequency was trained using the parameters of the previously trained model as the initial setting.

3. Results

a. Model performance

To test the performance of the CNN model in regions with different wind-wave climates, the data from three different locations were selected to train and validate three sets of CNN models, respectively. The previously mentioned point A is a point located in the Pacific Ocean swell pool, which can be used to test the feasibility of the model in the swell-dominated open ocean. The second selected point (point B) is located at 15°N , 115°E in the South China Sea, which reflects the performance of the model in (semi)enclosed seas. The third selected point (point C) is located at 50°S , 165°E in the westerlies, which reflects the performance of the model in areas impacted by strong prevailing winds. The locations of the three points are marked in Fig. 1. For each location, 30 parallel models for 30 frequencies were trained using the procedure in section 2c using the data from 1994 to 2013 and validated using the data from 2014 to 2018. Note that the grid structure (2 parameters $\times$ 5 direction sectors, 10 days $\times$ 8 fields day^{-1} , and 50 polar grid points in each 3° sector) is the same for the different points. This is to make sure that the model can be easily trained using the data from any given location in open oceans without changing the grid structure. However, note that for point B only a small fraction of the grids points in each sector is different from zero because of land blocking.

The model was trained and tested on a personal computer with a CPU of Intel Core i7-11700K and a GPU of NVIDIA GeForce RTX 3080-Ti. It took ~ 25 h to train the whole model (with 30 submodels of different frequencies) using 84 000 training samples, and it only takes a few seconds for the model to compute the results from 21 000 testing samples.

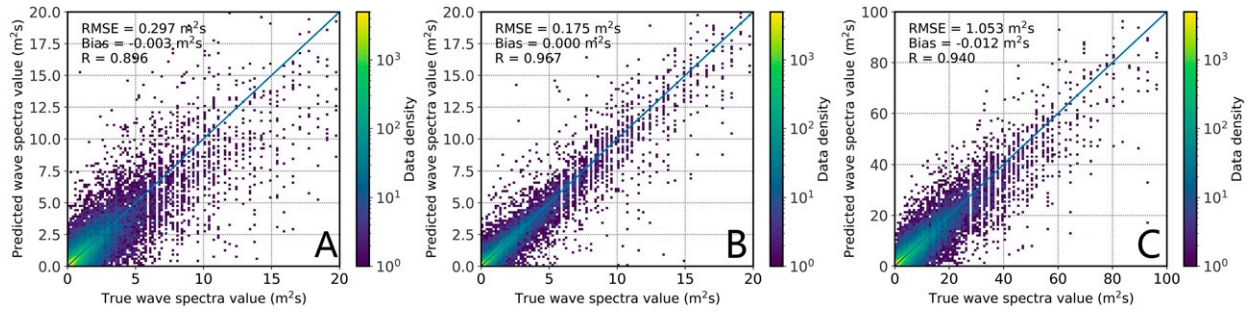


FIG. 4. Scatterplots for the comparison between true and predicted wave spectra on the test set, along with RMSE, bias, and R , for points (left) A, (center) B, and (right) C.

The bias, root-mean-square error (RMSE), and correlation coefficient (CC) were selected as the error metrics to evaluate the performance of the CNN model:

$$\text{bias} = \frac{1}{n} \sum_{i=1}^n (y_i - x_i), \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2}, \quad \text{and} \quad (4)$$

$$\text{CC} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (x_i - \bar{x})^2}}, \quad (5)$$

where x and y denote the spectral densities from ERA5 and our CNN models, respectively, and the bars over them denote their mean values. The scatterplots of spectral densities between ERA5 and the CNN models for the three points are shown in Fig. 4. The biases are close to zero and the CCs are ≥ 0.90 in all three points. Among the three cases, the model at point A and point B gives the lowest (0.90) and highest (0.97) CCs, respectively. The RMSE at point C is much higher than the other two points because of the much higher waves at this point, but the CC at point C (0.94) is higher than that at point A. A large proportion of the wave energy at point A comes from

the remote westerlies of both Hemispheres (Jiang and Mu 2019). After a storm occurs at the westerlies, the wind-input wave energy will go through complex physical processes, including breaking dissipation, nonlinear wave-wave interaction, frequency and angular spreading, and swell dissipation, before arriving at point A after ~ 10 -day propagation. Therefore, the prediction of swell arrival is to some extent an initial problem (e.g., Munk 1947; Pawka et al. 1984; Wingeart et al. 2001; Jiang et al. 2016), so it is more difficult to predict the remote swell energy than the local wind-sea energy. In contrast, point B is located at a semienlosed sea so that little swell energy from remote regions can propagate in, and the wind-sea spectra are more predictable. For point C, although it is often dominated by wind-sea, remote swells from other regions of the westerlies also play an important role.

Figure 5a shows the CCs of the CNN model on the validation sets as a function of frequency. The mean values of the spectral densities at each frequency at the three points are shown as a reference in Fig. 5b. The frequencies with the highest mean spectral densities for the three points are about 0.067, 0.108, and 0.081 Hz, respectively, which is in line with the wind-sea-swell dominance of the three points.

In general, the performance of CNN models at all three points becomes worse with the decrease in frequency. For all three points, the models have high accuracy in the high-frequency part, with CCs larger than 0.9 when the frequency is

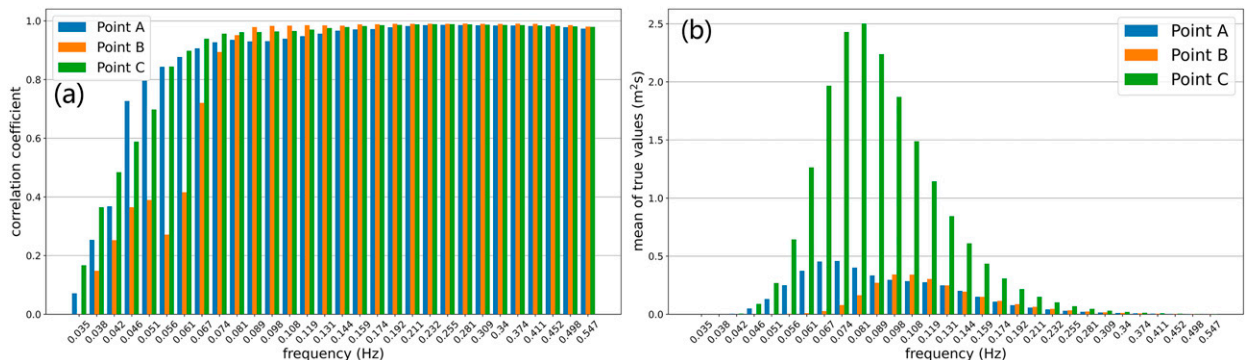


FIG. 5. (a) The correlation coefficients between CNN-estimated and ERA5 spectral densities for different frequencies on the validation sets at the three points (blue: point A, orange: point B, and green: point C). (b) The mean spectral densities at different frequencies at the three points.

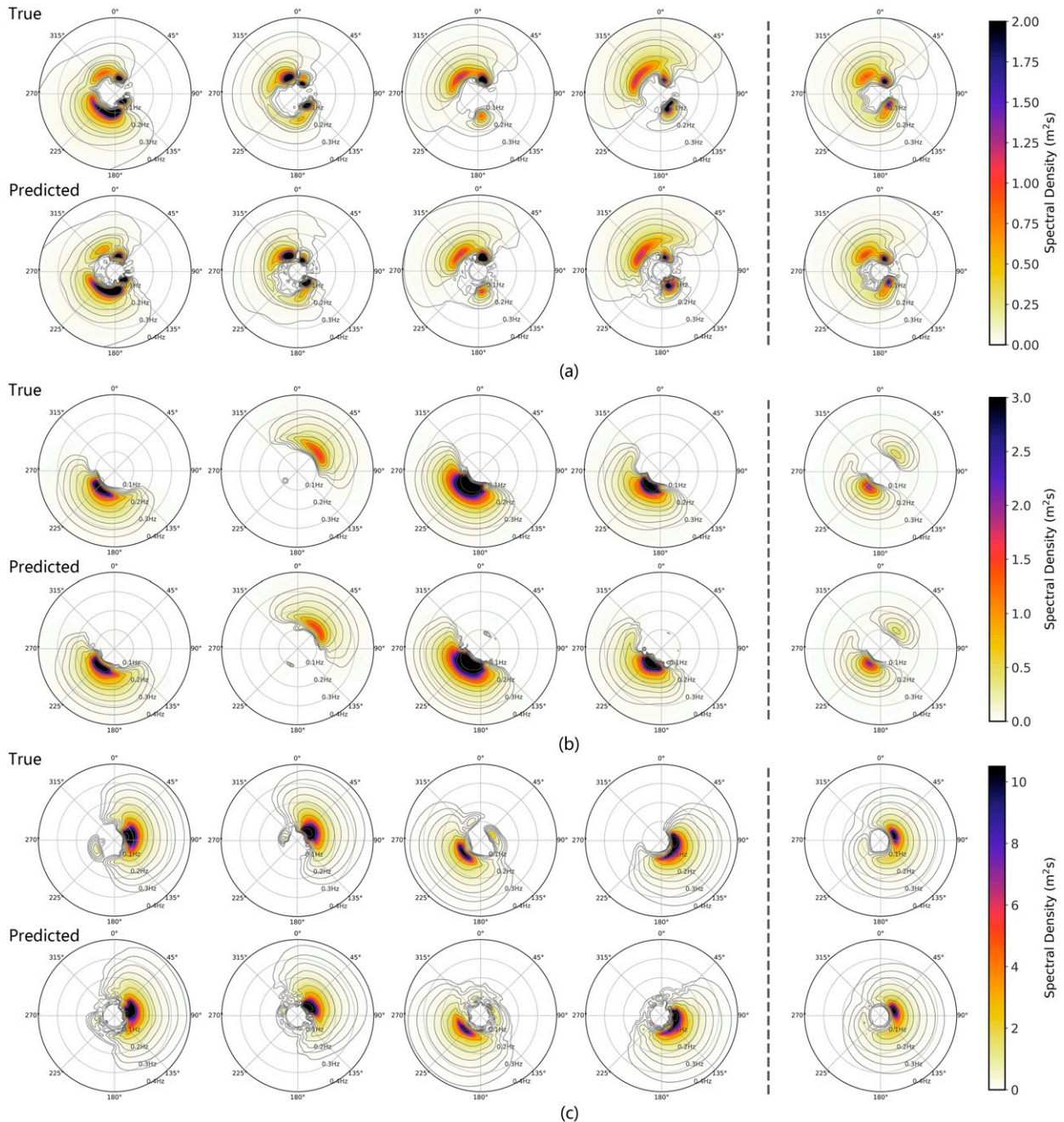


FIG. 6. The comparison of some randomly selected cases between ERA5 (top row of each subplot) and CNN-estimated (bottom row of each subplot) directional wave spectra for points (a) A, (b) B, and (c) C in 2016. The two spectra to the right of the dashed line of each subplot are the annual mean ERA5 (top of subplot) and CNN-estimated (bottom of subplot) directional wave spectra from the test sets in 2016. Note that the color bars are different for different subplots.

higher than 0.074 Hz. The wave energy in the high-frequency part is wind-sea-related and is generally in balance with the local wind (Phillips 1985), thus, it is relatively easy to predict. It is noted that the wave energy is negligible for frequencies lower than 0.046, 0.067, and 0.051 Hz at points A, B, and C, respectively. Therefore, the low CCs at frequencies lower than the three above number for the three points does not necessarily mean a

bad performance of the model. Figure 6 shows several arbitrarily selected cases for the comparison between ERA5 and CNN-estimated directional wave spectra (the full case-by-case comparison is given in the online supplemental material), as well as the comparisons of the corresponding annual mean spectra from the test sets at the three locations. As can be seen from the figure, the model gives a good estimate of the energy distribution of the

wave spectrum with respect to both hourly and annual mean results and, in particular, allows an accurate fit to regions with high energy density.

For point A, the shape of the CNN-estimated mean wave spectrum has a good agreement with the ERA5 one, with both the westerlies-generated and trade-wind-generated wave systems well shown. After checking many cases, we found that the main spectral peaks corresponding to these wave systems, both the local wind-seas with a wide directional spread and the narrow swells coming from the westerlies thousands of kilometers away, are also well predicted by the CNN model. This result shows the effectiveness of the model in the regions with a typical mixed wave state. Meanwhile, the CNN-estimated mean wave spectrum is slightly noisier than the ERA5 one, especially for the frequencies that are lower than 0.05 Hz, where some spurious peaks with very low wave energy are observable. From Fig. 5, it can also be seen that the CC becomes less than 0.8 when the frequency is lower than 0.046 Hz. From the randomly selected cases, it can also be seen that the agreement is good for high frequencies but that the spectrum shape becomes noisy and not as “smooth” as the ERA5 results when the frequency becomes lower. We checked many more cases, and all of them show this feature. There can be several potential reasons for these problems. First, the model is trained separately for different frequencies and each direction is a separate sample, resulting in the model not being able to identify some correlations between neighboring frequency and direction bins. Second, some numerical reasons, such as the GSE (Tolman 2002; it refers to the phenomenon in NWMs that swell propagation at coarse spectral resolution leads to the disintegration of continuous swell fields into discrete ones), can induce “random” numerical errors to the ERA5 modeled spectra at low frequencies, making the CNN difficult to capture the features at low-frequency spectra. Third, the formation of these swell systems involves many complex physical processes such as dissipation, diffraction, frequency dispersion, angular spreading, and wave-wave interaction, which is also more difficult for the CNN to “learn.”

In general, the agreement between ERA5 and CNN spectra is better at point B than at point A. For point B, because of the limited fetch, the waves at this point are mainly wind-sea and locally generated young swells that are northeastward or southwestward. These waves are simulated well by the CNN and with respect to both spectral shape and energy level. The swells generated at far fields from the western Pacific Ocean can propagate into the South China Sea through the Luzon Strait (Qian et al. 2020). Such weak swell systems can also be predicted using the CNN model (e.g., the system at $\sim 225^\circ$ direction in the second column of Fig. 6b), indicating the model can also give reasonable outputs at this location. The wind projections to point B are all zeros at far fields due to the land blocking, except for the direction pointing to the Luzon Strait, which reduced the size of the “effective” input of the CNN model. Also, the limited fetch reduced the complexity of processes of swell propagation. These factors make the CNN model very suitable for the prediction of waves at (semi)enclosed seas.

The performance of the CNN model at point C is similar to that at point A. The agreement is good between the two mean wave spectra. The main spectral peaks, which usually correspond to the waves generated by the local westerlies, are well modeled by the CNN model. However, for the low-frequency waves with relatively low spectral densities that propagate along a different direction from the main wave system, the CNN gives much noisier outputs. Although most partitions in the ERA5 outputs are found in the CNN-estimated spectra, many small but spurious partitions are also observable. Some smoothing might be needed before using these spectra. The reasons for the noise in the spectra at point C can also be attributed to the large propagation distance of the low-frequency swells and the complex and strong nonlinear wave-wave interaction in the midlatitudes storms. Also, the Antarctic sea ice, which is not considered in our CNN model, might have some impact on the wave energy, especially for those with a northward component (impacted by the polar easterlies). However, it is also noted that the energy level of these spurious peaks is much lower than that of the main wave partitions. Therefore, the spectra at this point derived from the CNN can still be practical for many applications.

To further test the validity of the results from a statistical point of view, the integral wave parameters, including SWH, MWP, and MWD (the definitions of these parameters are given in the appendix) computed from the ERA5 and CNN-estimated directional wave spectra are also compared with each other. It is noted that two spectra with different shapes can sometimes have the same integral wave parameters. However, Fig. 6 has shown the agreement between the spectral shapes of the ERA5 and CNN-estimated results. In this case, the comparison of integral wave parameters can serve as an indirect way to evaluate our CNN model. Such an indirect method is widely used in the verification of numerical wave models where SWHs from spaceborne altimeters are often used to evaluate the performance of a model (e.g., Alday et al. 2021; Liu et al. 2021).

The comparisons of integral wave parameters between ERA5 and CNN-estimated results are shown in Fig. 7 as scatterplots. In general, the results show that the overall results of the three integral wave parameters are good in the three locations with different wave conditions, which confirms the validity of the CNN model's prediction. The errors are similar to those between contemporary global NWM hindcasts and observations (e.g., Liu et al. 2021; Alday et al. 2021). The overall biases are all small for the three parameters at the three points. However, a significant bias is found for MWP < 5 s at point B, which leads to an overall bias of 0.12 s. When the MWP is less than 5 s, the sea-state is dominated by high-frequency wind-sea but with low energy. In this case, a very small noise at low frequencies induced by the CNN model (swells will not exist in the open ocean for such a low MWP) can lead to a significant overestimation of MWP. With respect to RMSEs and CCs of SWH, the agreement is the best at point B where the CC of SWH can reach 0.99, showing that such a CNN model can be very suitable for (semi)enclosed seas. The CC of SWH is relatively low at point A (only 0.91), because the SWH at this point generally varies in a small range

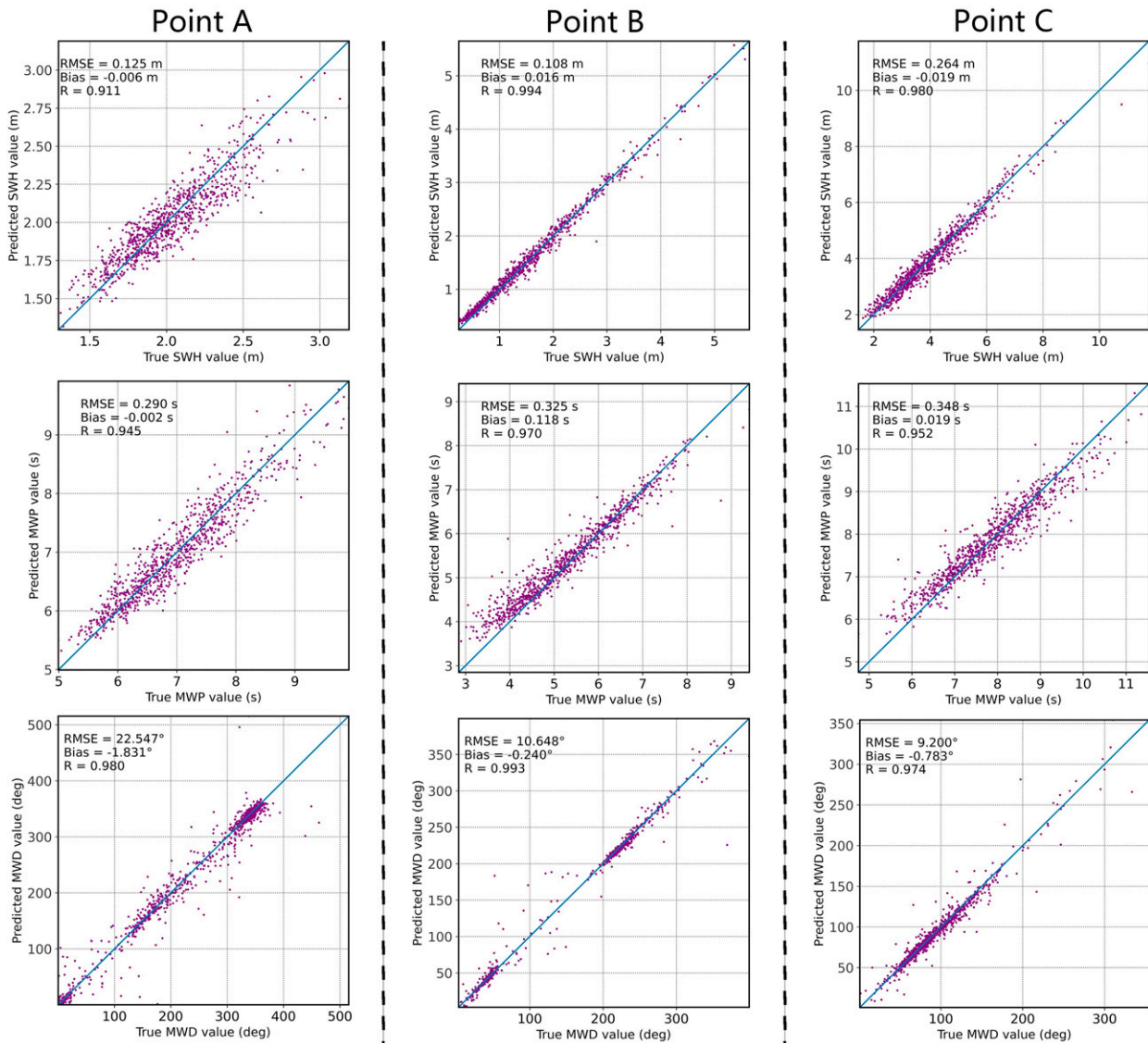


FIG. 7. Scatterplots of (top) SWH, (middle) MWP, and (bottom) MWD computed from the ERA5 and CNN-estimated directional wave spectra for points (left) A, (center) B, and (right) C.

(1.2–3.2 m for point A but 0–5.5 m for point B). For MWP and MWD, the RMSEs and CCs are respectively low and high at all three points, but outliers are common in the scatterplots of MWD, especially at point A. This is partly due to the “180° ambiguity” of computing MWD when two wave systems have similar energy but nearly opposite propagation directions, which is common at point A (e.g., the waves generated by the trade winds and westerlies of the Northern and Southern Hemisphere, respectively). Another reason is that small noise of the CNN output can also have a large impact on the estimation of MWD when the overall wave energy is low.

b. A generic CNN model for different locations

In the above method, different sets of CNNs were trained from different locations in the open ocean, and this method

works well. The models in the three different locations are established based on the same physical mechanisms and the same model structure. Therefore, it seems to be possible to establish a data-driven CNN model adaptive to different locations in the open ocean after some training. Therefore, in this section, we tried to establish a generic CNN model that can be used to predict directional wave spectra at different locations in the ocean.

One might think that a CNN model can also be trained directly using the wind input in a regular $360^\circ \times 180^\circ$ global grid. However, such a regular grid will have a much larger size of input vectors, so it will be much more difficult for the training process to converge, and more data are needed to train. This problem is significantly reduced using the polar grid (1/24 input vector size and 24 times samples, as mentioned

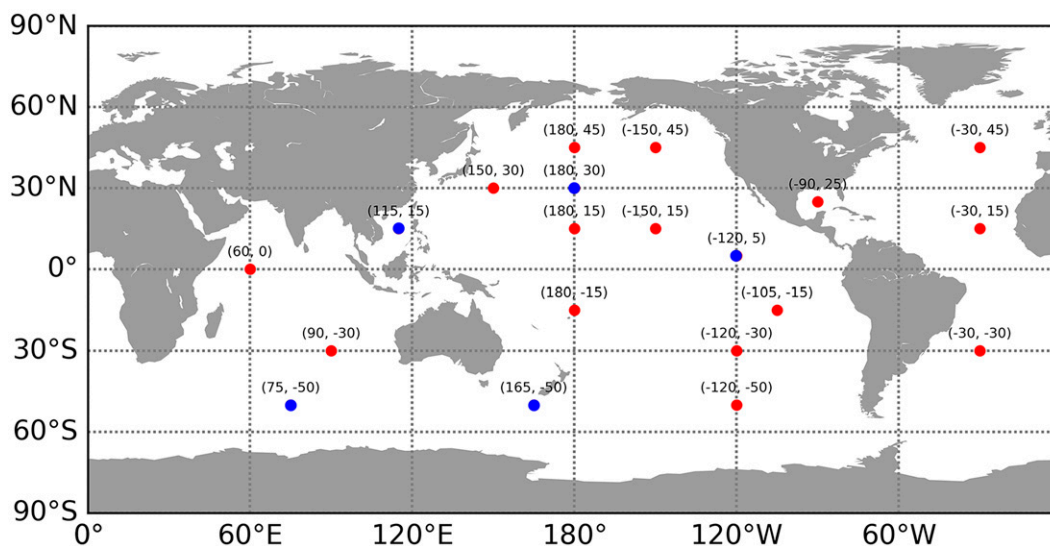


FIG. 8. The locations of the 20 points used to train and test the generic CNN models. Points used for training are marked in red, and points used for testing are marked in blue. The coordinates are in the format (longitude, latitude), where east and north are positive and west and south are negative.

previously). Another advantage of such a polar grid centered on the target point is its symmetry in different directions. If the surrogate CNN model in a polar grid really “learns” the physical processes with respect to the generation and propagation of waves from the data, such a CNN model should be able to predict the wave at other locations as the physics behind the evolution of waves are the same for different locations (in the deep ocean). For a regular grid (unless it is also reorganized to be centered on the target point), the CNN model for one point cannot be used at other locations.

To establish a generic CNN model that is adaptive for different locations, the new CNN model was trained with samples from 15 different points and tested with samples from 5 points (Fig. 8) in the open ocean. The locations of these points cover the westerlies and trade wind zones in the three oceans as well as semienclosed seas, and include regions of wind-sea-dominated, swell-dominated, and typical mixed seas. Only open ocean data are used here because bathymetry and current information are necessary for the prediction of wave spectra in coastal regions. For each point, the wind data were also reorganized into the polar grid centered on itself. Then, each input sample will also be 10 days’ wind information in a 15° sector, and the corresponding output sample will be the spectral densities in the corresponding direction at this location. This generates a training set containing 63 000 samples from 15 different locations and a test set containing 21 000 samples from 5 other different locations using the data from the year 2016. The details of the CNN model are the same as those described in section 2c. We also tried to increase the numbers of layers, kernels, and neurons of the CNN, which does not give better results.

The overall CCs of the CNN model’s output for different frequencies on the validation set are shown in Fig. 9. Similar to the results in Fig. 5a, the CCs increase with the increase of

frequency, and the value becomes larger than 0.8 or 0.9 when the frequency is higher than 0.061 or 0.074 Hz, respectively. These results are slightly worse than the results in Fig. 5a, and the reasons for the relatively low accuracy at the low-frequency part are similar to those for the single-point CNN model. To further evaluate the model’s performance, the scatterplots of spectral densities, SWHs, MWP, and MWDs between ERA5 and the CNN model in the validation set are shown in Fig. 10. The comparison of spectral density shows that the high-energy spectral points are often underestimated. The CC shown in Fig. 10a is 0.90, which is lower than two of the three sets of results (points B and C) in Fig. 4 and about the same as one of the sets (point A), showing that it is more difficult for the model to “learn” from the data when the location is not fixed. However, regarding the integral wave parameters, the generic CNN model can still have reasonably good accuracy. The RMSEs for SWH, MWP, and MWD are 0.29 m, 0.46 s, and 21°, respectively, and the corresponding CCs are 0.99, 0.96, and 0.98, which can be regarded as a good agreement. Some outliers exist in the MWD, partly due to the aforementioned “180° ambiguity,” especially when SWH is low.

We also made visual checks on many selected cases for the comparison between ERA5 and CNN-estimated directional wave spectra. Because of the limited space, we still used the twelve cases at the three points in Fig. 6 as examples, and the results are shown in Fig. 11. Still, the model can give a reasonable estimate of the energy distribution of the wave spectrum. The main peaks with significant wave energy are all well predicted by the CNN model, and most less significant peaks are also visible in the CNN-predicted spectra. The directional wave spectra from the generic CNN model are slightly noisier than those from the single-location model.

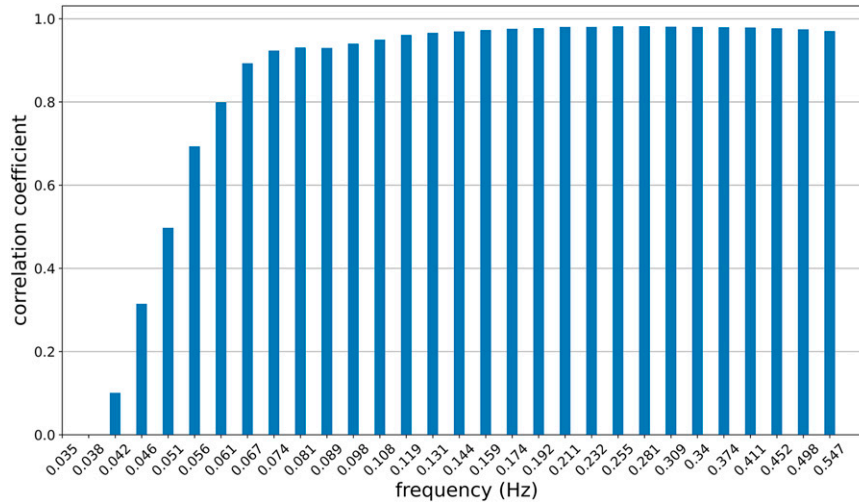


FIG. 9. The correlation coefficients between spectral densities estimated from multilocation generic CNN model and those from ERA5 for different frequencies on the validation set.

For point A, the relatively high noise level for the low-frequency part in Fig. 6 still exists and becomes slightly higher in the corresponding results in Fig. 11. However, both the wide peak corresponding to the trade wind-generated wind-sea and the narrow peaks corresponding to the westerlies-generated swells are still well shown in the CNN results, with respect to both single and annual mean directional wave spectra. This shows the effectiveness of such a generic CNN model in the regions with a typical mixed wave state.

For point B, the higher noise level in the low-frequency part in the generic CNN model is more visible. In the directions opposite to main peaks, the energy level should be close to zero except for the aforementioned intruding swells from Luzon Strait (second column of Figs. 6b and 11b). However, the generic CNN model gives clear noise in these directions, and the level of these noises is comparable to the energy of the weak intruding swells, making the intruding swells almost not identifiable in the CNN-simulated results. The noise in the low-frequency part also has an observable impact on the annual mean spectrum. Fortunately, the energy level of these intruding swells does not have impacts on most engineering applications. It is noted that such low-frequency wave systems with low energy levels are often not identifiable in the in situ

measurement (Jiang et al. 2022). The shape and energy of the main northeastward or southwestward wind-sea spectral peak are still simulated well by the CNN.

For point C, the performance of the generic CNN model is similar to that at point B. The main wind-sea spectral peaks with the highest energy are well simulated. Most peaks of the most swell systems with significant energy are also observable, for example, the peaks at $\sim 250^\circ$ in the first column, $\sim 290^\circ$ in the second column, and $\sim 70^\circ$ and $\sim 340^\circ$ in the third column of Fig. 11c. The noise also leads to some less significant but observable spurious peak at low frequencies. However, the energy levels of these spurious peaks are lower than main partitions, making the directional wave spectra from the generic CNN model still practical for many applications.

4. Discussion

The results at the three points all show that the CNN model is better at simulated wind-seas than swells. This is because the simulation of local wind-sea spectra can be regarded as a forcing response problem while the simulation of remote swell spectra becomes a more complicated combination of a forcing response problem (the generation of waves) and an

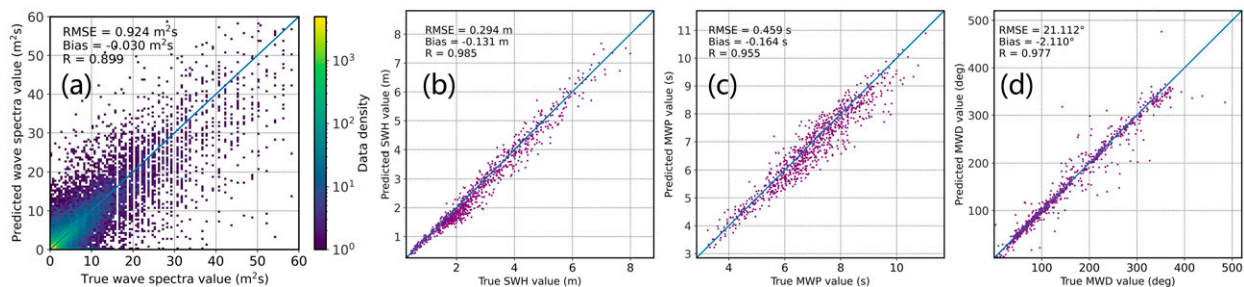


FIG. 10. (a) As in Fig. 4, but for the generic CNN model for five different locations in the validation set. (b)–(d) Similar to Fig. 7, but also for the generic CNN model for five different locations in the validation set.

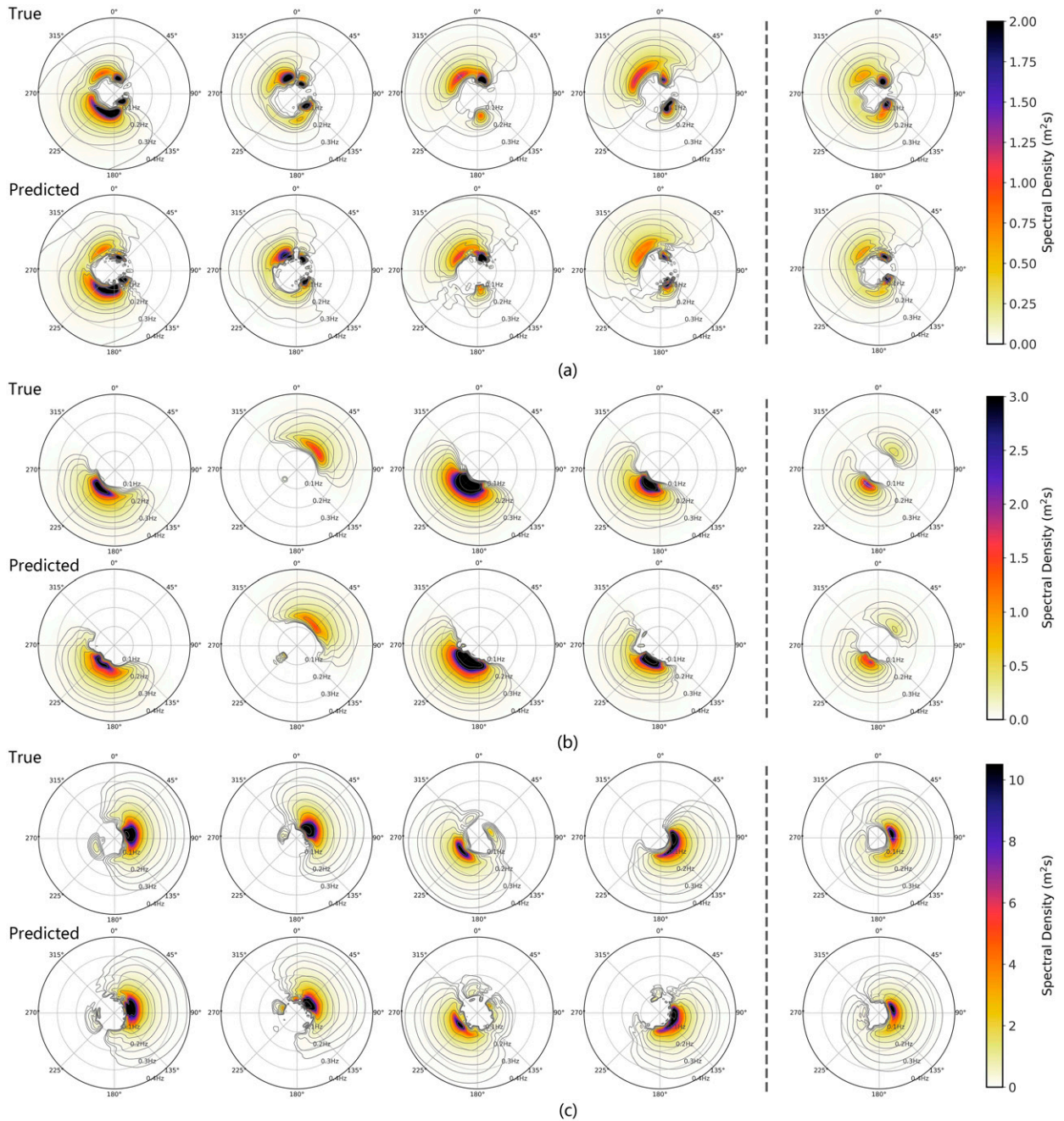


FIG. 11. As in Fig. 6, but the predicted results are obtained using the generic CNN model for different locations. Note that no data from the three points shown in the figure were used for the training of the generic CNN model.

initial value problem (large-distance propagation of waves). It is also more difficult for the numerical wave models to accurately predict swells than wind-seas (e.g., Jiang et al. 2016).

Figure 12 shows the CCs between U_{proj}^2 and spectral densities at 0.281, 0.108, and 0.042 Hz with different spatial-temporal distances to point A. Here we selected U_{proj}^2 instead of U_{proj} because it is better correlated to the spectral densities at all frequencies. The 24-h periodical strips are due to the diurnal

variability of wind. For the high-frequency part, the spectral densities are strongly correlated to the present local wind (Fig. 12a shows the highest CC of 0.94). This is because such a high frequency is usually in the tail of the spectrum, which responds rapidly to the change in wind conditions. It will be very simple for the CNN to learn such a quantitative relation from the data. In fact, the structure of the CNN model can be much simpler and the input spatial-temporal range

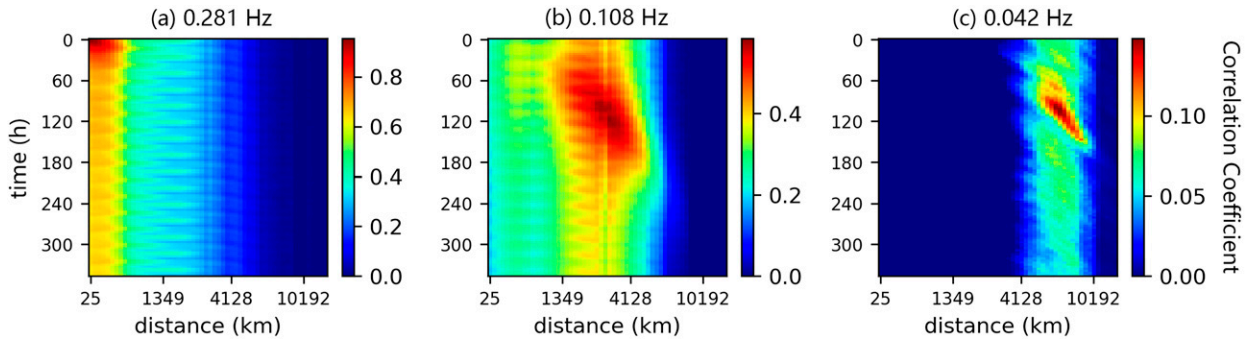


FIG. 12. Correlation coefficients between U_{proj}^2 and spectral densities at (a) 0.281, (b) 0.108, and (c) 0.042 Hz with different spatial–temporal distances to point A.

can be much smaller if one only wants to model the spectral tails.

The highest CC in Fig. 12b is found in the range of 2000–4000 km and 60–150 h, with the highest value of less than 0.6. The evolution of spectra energy in such intermediate frequencies becomes more complex than the condition at high frequencies. The spectral density at these frequencies is dominated by the nonlinear quadruplet wave–wave interaction, which keeps pumping the wind-input high-frequency energy into lower frequencies. In this case, the spectral densities here are the result of the integral effect of wind along the wave propagation direction, or to say, the growth of waves. Empirical relations have been established such as the famous JONSWAP spectrum (Hasselmann et al. 1973), and such empirical relations are the bases of second-generation numerical wave models. A CNN model can also relearn such empirical relations from the data. One potential advantage of the CNN model is that its ability to fit the complex nonlinear relationship between input and output might enable it to relearn such complex relations with varying space–time winds. The CC between the CNN-estimated and ERA5 spectral densities at 0.108 Hz is more than 0.93 at point A (Fig. 5a), which is much higher than the highest value in Fig. 12b. This implies that the CNN model has probably succeeded in learning such relations.

For low frequencies, the condition becomes even more complicated. The energy at these frequencies was first generated through the process of nonlinear quadruplet wave–wave interaction in the far field, and then propagates out of the generation area with a very high speed (the group speed of 0.042-Hz waves will be $\sim 19 \text{ m s}^{-1}$). During the propagation of waves, the energy was then attenuated by the frequency dispersion, angular spreading, and nonbreaking dissipation. All these effects need to be relearned by the CNN only from the input historical far-field wind and output local spectral density, which makes it a very challenging task. However, the CNN seems to at least capture some of these processes. The CC between the CNN-estimated and ERA5 spectral densities at 0.042 Hz is ~ 0.65 , which is much higher than the weak but clear peak in Fig. 12c with a CC of ~ 0.15 . Meanwhile, it is noted that the spectral densities at such a low frequency in NWMs including ERA5 are not very reliable. They are the most impacted by the numerical errors induced by the GSE and discrete interaction approximation, and such numerical errors will also impact the training of CNN at low frequencies.

4. Conclusions

Directional wave spectra are important for many ocean-related practical applications. Previous studies show that the wave spectral densities at a given point in the open ocean are correlated with the local and remote wind field. In this study, an empirical model of predicting the directional wave spectra based on deep learning is proposed. Two types of CNN models were trained, one for a single given location and one adaptive for any locations in the global open ocean (the generic CNN model), both using the reorganized wind field in a polar grid as the input.

Using three locations with different spectral wave climate properties, some with several wave systems coexisting, the performances of the two types of CNN models were evaluated. Each of them has its advantages and disadvantages. The performance of the CNN model that is adaptive for different locations in the open ocean is slightly worse than the single-point model, with a higher noise level in the low-frequency part of the spectra. However, such a generic CNN model can be used for almost any location in the open ocean even under the condition of training data being not available, which is the target of being “generic.” In general, both types of models are effective in simulating directional wave spectra: the spectral densities of almost all significant spectral peaks and the spectral shape of the local wind-sea peaks are well predicted. The integral wave parameters from the spectra also have a good agreement with the numerical wave models. Such CNN models allow for the prediction of the wave spectra at a given location with low computational cost and can be helpful for the fast modeling of the wave spectrum at a given location and the study of local spectral wave climate.

We have to admit that there is a large space to improve such a kind of empirical spectral wave model. The largest problem in the current model is the relatively poor performance in predicting the swell systems with low energy at very low frequencies ($< 0.05 \text{ Hz}$). The prediction of remotely generated long swells is much more complex than that of local wind-seas due to the fast wave propagation of large distances. Also, the model is not applicable yet in coastal regions because the bathymetry and current information are not used as inputs at this stage. However, this study shows the feasibility of using local and remote wind fields to empirically predict directional wave spectra of different wind-sea and swell partitions,

respectively. Therefore, future work can focus on the optimization of the model using more advanced deep learning frameworks. For example, one can consider capturing the spatial-temporal relevance of the wind input by introducing an attention mechanism into the model or capturing the features in different scales using methods such as U-Net. In addition, when using these advanced deep learning frameworks, it is possible to take into account other forces such as currents. Another possible direction is to add some physical constraints to the output spectra using some regularization terms in the loss function, which makes the model a physics-informed neural network. At this stage, however, we did not figure out clearly how to add these physics restrictions. The final goal of such a kind of AI model is to establish an accurate empirical spectral wave model. With the fast development of AI, it is promising that this goal may be achieved.

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Data availability statement. The ERA5 data are available at the Climate Data Store (<https://cds.climate.copernicus.eu/>).

APPENDIX

Definitions of SWH, MWP, and MWD

The SWH, MWP, and MWD are computed as follows:

$$\text{SWH} = 4.04\sqrt{m_0}, \quad (\text{A1})$$

$$\text{MWP} = m_0/m_2, \quad \text{and} \quad (\text{A2})$$

$$\text{MWD} = \arctan(\text{SF}/\text{CF}), \quad (\text{A3})$$

where

$$m_i = \iint f^i S(f, \theta) df d\theta, \quad (\text{A4})$$

$$\text{SF} = \iint \sin \theta \times S(f, \theta) df d\theta, \quad (\text{A5})$$

$$\text{CF} = \iint \cos \theta \times S(f, \theta) df d\theta, \quad (\text{A6})$$

and $S(f, \theta)$ is the spectral density at frequency f and direction θ .

REFERENCES

- Alday, M., M. Accensi, F. Ardhuin, and G. Dodet, 2021: A global wave parameter database for geophysical applications. Part 3: Improved forcing and spectral resolution. *Ocean Modell.*, **166**, 101848, <https://doi.org/10.1016/j.ocemod.2021.101848>.
- Ardhuin, F., B. Chapron, and F. Collard, 2009: Observation of swell dissipation across oceans. *Geophys. Res. Lett.*, **36**, L06607, <https://doi.org/10.1029/2008GL037030>.
- Barber, N. F., and F. Ursell, 1948: The generation and propagation of ocean waves and swell. Part I. Wave periods and velocities. *Philos. Trans. Roy. Soc.*, **A240**, 527–560, <https://doi.org/10.1098/rsta.1948.0005>.
- Bengio, Y., Y. Lecun, and G. Hinton, 2021: Deep learning for AI. *Commun. ACM*, **64**, 58–65, <https://doi.org/10.1145/3448250>.
- Booij, N., L. H. Holthuijsen, and R. C. Ris, 1997: The “swan” wave model for shallow water. *Coastal Engineering* 1996, B. L. Edge, Ed., ASCE, 668–676, <https://doi.org/10.1061/9780784402429.053>.
- Cagigal, L., A. Rueda, A. Ricondo, J. Pérez, N. Ripoll, G. Coco, and F. J. Méndez, 2021: Climate-based emulator of distant swell trains and local seas approaching a Pacific atoll. *J. Geophys. Res. Oceans*, **126**, e2020JC016919, <https://doi.org/10.1029/2020JC016919>.
- Chen, G., B. Chapron, R. Ezraty, and D. Vandemark, 2002: A global view of swell and wind sea climate in the ocean by satellite altimeter and scatterometer. *J. Atmos. Oceanic Technol.*, **19**, 1849–1859, [https://doi.org/10.1175/1520-0426\(2002\)019<1849:AGVOSA>2.0.CO;2](https://doi.org/10.1175/1520-0426(2002)019<1849:AGVOSA>2.0.CO;2).
- Earle, M. D., K. E. Steele, and D. W. C. Wang, 1999: Use of advanced directional wave spectra analysis methods. *Ocean Eng.*, **26**, 1421–1434, [https://doi.org/10.1016/S0029-8018\(99\)00010-4](https://doi.org/10.1016/S0029-8018(99)00010-4).
- Espejo, A., P. Camus, I. J. Losada, and F. J. Méndez, 2014: Spectral ocean wave climate variability based on atmospheric circulation patterns. *J. Phys. Oceanogr.*, **44**, 2139–2152, <https://doi.org/10.1175/JPO-D-13-0276.1>.
- Ham, Y.-G., J.-H. Kim, and J.-J. Luo, 2019: Deep learning for multi-year ENSO forecasts. *Nature*, **573**, 568–572, <https://doi.org/10.1038/s41586-019-1559-7>.
- Hasselmann, K., and Coauthors, 1973: Measurements of wind-wave growth and swell decay during the Joint North Sea Wave Project (JONSWAP). *Dtsch. Hydrogr. Z.*, **8** (Suppl. A), 1–93.
- Hasselmann, S., K. Hasselmann, J. H. Allender, and T. P. Barnett, 1985: Computations and parameterizations of the nonlinear energy transfer in a gravity-wave spectrum. Part II: Parameterizations of the nonlinear energy transfer for application in wave models. *J. Phys. Oceanogr.*, **15**, 1378–1391, [https://doi.org/10.1175/1520-0485\(1985\)015<1378:CAPOTN>2.0.CO;2](https://doi.org/10.1175/1520-0485(1985)015<1378:CAPOTN>2.0.CO;2).
- Hegermiller, C. A., A. Rueda, L. H. Erikson, P. L. Barnard, J. A. A. Antolinez, and F. J. Mendez, 2017: Controls of multimodal wave conditions in a complex coastal setting. *Geophys. Res. Lett.*, **44**, 12 312–12 323, <https://doi.org/10.1002/2017GL075272>.
- Hersbach, H., and Coauthors, 2020: The ERA5 global reanalysis. *Quart. J. Roy. Meteor. Soc.*, **146**, 1999–2049, <https://doi.org/10.1002/qj.3803>.
- Holthuijsen, L. H., 2007: *Waves in Oceanic and Coastal Waters*. Cambridge University Press, 405 pp.
- James, S. C., Y. Zhang, and F. O'Donncha, 2018: A machine learning framework to forecast wave conditions. *Coast. Eng.*, **137**, 1–10, <https://doi.org/10.1016/j.coastaleng.2018.03.004>.
- Jiang, H., 2022: Wind speed and direction estimation from wave spectra using deep learning. *Atmos. Meas. Tech.*, **15**, 1–9, <https://doi.org/10.5194/amt-15-1-2022>.
- , and L. Mu, 2019: Wave climate from spectra and its connections with local and remote wind climate. *J. Phys. Oceanogr.*, **49**, 543–559, <https://doi.org/10.1175/JPO-D-18-0149.1>.

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- , A. V. Babanin, and G. Chen, 2016: Event-based validation of swell arrival time. *J. Phys. Oceanogr.*, **46**, 3563–3569, <https://doi.org/10.1175/JPO-D-16-0208.1>.
- , A. Mouche, H. Wang, A. V. Babanin, B. Chapron, and G. Chen, 2017: Limitation of SAR quasi-linear inversion data on swell climate: An example of global crossing swells. *Remote Sens.*, **9**, 107, <https://doi.org/10.3390/rs9020107>.
- , A. Mironov, L. Ren, A. V. Babanin, J. Wang, and L. Mu, 2022: Validation of wave spectral partitions from SWIM instrument on-board CFOSAT against *in situ* data. *IEEE Trans. Geosci. Remote Sens.*, **60**, 1–13, <https://doi.org/10.1109/TGRS.2021.3110952>.
- LeCun, Y., Y. Bengio, and G. Hinton, 2015: Deep learning. *Nature*, **521**, 436–444, <https://doi.org/10.1038/nature14539>.
- Liu, Q., and Coauthors, 2021: Global wave hindcasts using the observation-based source terms: Description and validation. *J. Adv. Model. Earth Syst.*, **13**, e2021MS002493, <https://doi.org/10.1029/2021MS002493>.
- Munk, W. H., 1947: Tracking storms by forerunners of swell. *J. Atmos. Sci.*, **4**, 45–57, [https://doi.org/10.1175/1520-0469\(1947\)004<0045:TSBFOS>2.0.CO;2](https://doi.org/10.1175/1520-0469(1947)004<0045:TSBFOS>2.0.CO;2).
- O'Donncha, F., Y. Zhang, B. Chen, and S. C. James, 2018: An integrated framework that combines machine learning and numerical models to improve wave-condition forecasts. *J. Mar. Syst.*, **186**, 29–36, <https://doi.org/10.1016/j.jmarsys.2018.05.006>.
- Pawka, S. S., D. L. Inman, and R. T. Guza, 1984: Island sheltering of surface gravity waves: Model and experiment. *Cont. Shelf Res.*, **3**, 35–53, [https://doi.org/10.1016/0278-4343\(84\)90042-6](https://doi.org/10.1016/0278-4343(84)90042-6).
- Pérez, J., F. J. Méndez, M. Menéndez, and I. J. Losada, 2014: ESTELA: A method for evaluating the source and travel time of the wave energy reaching a local area. *Ocean Dyn.*, **64**, 1181–1191, <https://doi.org/10.1007/s10236-014-0740-7>.
- Phillips, O. M., 1985: Spectral and statistical properties of the equilibrium range in wind-generated gravity waves. *J. Fluid Mech.*, **156**, 505–531, <https://doi.org/10.1017/S0022112085002221>.
- Qian, C., H. Jiang, X. Wang, and G. Chen, 2020: Climatology of wind-seas and swells in the China seas from wave hindcast. *J. Ocean Univ. China*, **19**, 90–100, <https://doi.org/10.1007/s11802-020-3924-4>.
- Rueda, A., and Coauthors, 2017: Multiscale climate emulator of multimodal wave spectra: MUSCLE-spectra. *J. Geophys. Res. Oceans*, **122**, 1400–1415, <https://doi.org/10.1002/2016JC011957>.
- Snodgrass, F. E., K. F. Hasselmann, G. R. Miller, W. H. Munk, and W. H. Powers, 1966: Propagation of ocean swell across the Pacific. *Philos. Trans. Roy. Soc.*, **A249**, 431–497, <https://doi.org/10.1098/rsta.1966.0022>.
- Sverdrup, H. U., and W. H. Munk, 1946: Theoretical and empirical relations in forecasting breakers and surf. *Eos*, **27**, 828–836, <https://doi.org/10.1029/TR027i006p00828>.
- The WISE Group, 2007: Wave modelling: The state of the art. *Prog. Oceanogr.*, **75**, 603–674, <https://doi.org/10.1016/j.pocean.2007.05.005>.
- Tolman, H. L., 1991: A third-generation model for wind waves on slowly varying, unsteady and inhomogeneous depths and currents. *J. Phys. Oceanogr.*, **21**, 782–797, [https://doi.org/10.1175/1520-0485\(1991\)021<0782:ATGMFW>2.0.CO;2](https://doi.org/10.1175/1520-0485(1991)021<0782:ATGMFW>2.0.CO;2).
- , 2002: Alleviating the garden sprinkler effect in wind wave models. *Ocean Modell.*, **4**, 269–289, [https://doi.org/10.1016/S1463-5003\(02\)00004-5](https://doi.org/10.1016/S1463-5003(02)00004-5).
- , and WAVEWATCH III Development Group, 2014: User manual and system documentation of WAVEWATCH III version 4.18. MMAB Tech. Note 316, 311 pp., <https://polar.ncep.noaa.gov/waves/wavewatch/manual.v4.18.pdf>.
- Wingert, K. M., T. H. C. Herbers, W. C. O'Reilly, P. A. Wittmann, R. E. Jensen, and H. L. Tolman, 2001: Validation of operational global wave prediction models with spectral buoy data. *Ocean Wave Measurement and Analysis 2001*, B. L. Edge and J. M. Hemsley, Eds., ASCE, 590–599, [https://doi.org/10.1061/40604\(273\)61](https://doi.org/10.1061/40604(273)61).
- Zheng, G., X. Li, R.-H. Zhang, and B. Liu, 2020: Purely satellite data-driven deep learning forecast of complicated tropical instability waves. *Sci. Adv.*, **6**, eaba1482, <https://doi.org/10.1126/sciadv.aba1482>.